



Department of Electronics and Communication Engineering Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E.ECE B

Course Code & Title: EC8751 Optical Communication

Name of the Faculty member (s): Mrs.R.Ramalakshmi

Innovative Practice Description

- **Unit / Topic:** Unit III / Laser structures and radiation patterns, Single mode laser, External modulation-temperature effort
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 9
- **Activity Chosen:** In Class Teams
- **Justification:**

The students will be able to understand the concepts of Laser structures and radiation patterns, Single mode laser, External modulation-temperature effort with the help of In Class Teams activity more easily. The students will be able to easily remember the concepts if they share the concepts with the team members.

- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

In Class Teams is a collaborative teaching strategy. It can be used to help students form individual ideas, discuss and share with the others in-group. It can be used before reading or teaching a concept and works better with smaller groups.

The students were in the class room. First the students were taught with the concepts of Laser structures and radiation patterns, Single mode laser, External modulation-temperature effort. Then the students were allowed to discuss the topics among their friends. Then two students grouped and shared their views on the topic. Since the students discussed among their friends, they understood the concepts more clearly.

- **CO – PO / PSO mapping:**

CO	PO1	PO3	PO9	PO10	PSO1	PSO3
CO3	3	3	2	2	3	3

(1 – Low

2 – Moderate

3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9	PO10	PSO3
	2	2	3
Justification for correlation	In-Class Teams activity helps the students to present results as a team, with smooth integration of contributions from all individual efforts.	The topic deals with optical sources like LED, Laser to find internal and external quantum efficiency. This task is accomplished through team activity.	To analyze the various optical sources, laser structures, radiation patterns & External modulation-temperature effort, classify various types of detectors and to design & compare optical detectors, this activity is really useful.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

The effectiveness of the activity was felt when the students were asked the questions on the topic, they were able to answer easily. They also understood the concepts more easily and able to remember while writing the test.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The students were more interested in learning the concepts if they are sharing the concepts through their friends. They clarified their doubts with the neighboring students and the faculty. They learnt the concepts and asked more doubts in the topic and learnt many things. Outcome attainment has also been increased since the students' performance in the test is also good.

❖ ***Challenges faced in implementation:***

Basically some time is needed to explain the basic concepts of laser structures, radiation patterns to the students and also to discuss among the students.

References:

Textbooks:

1. P Chakrabarti, "Optical Fiber Communication, McGraw Hill Education (India) Private Limited, 2016 (UNIT I, II, III)
2. Gerd Keiser, "Optical Fiber communication" MC Graw -Hill international, 4th Edition, 2010

Other References:

Books / Website/ Journal/e-books/ Videos/ NPTEL/ Matlab Online/ Online Course etc.

1. John M. Senior, "Optical Fiber communication", Second Edition, Pearson Education, 2007
2. Ramaswami, Sivarajan and sasaki "Optical Networks ", Morgan kaufmann, 2009
3. J. Senior, "optical communication, principles and practice", Prentice Hall of India, 3rd Edition, 2008.
4. NPTEL video on "Advanced Optical Communication"
Link: <https://nptel.ac.in/courses/117/101/117101002/>



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Name of the Faculty member (s): Mrs.R.Ramalakshmi

Innovative Practice Description

- **Unit / Topic:** Unit V / SONET/SDH Rings and Networks
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 14
- **Activity Chosen:** Reciprocal Peer Questioning
- **Justification:**

The students will be able to understand the concepts of SONET/SDH Rings and Networks with the help of Reciprocal Peer Questioning activity more easily. The students will be able to easily remember the concepts if questions are being asked to them.

- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

Reciprocal Peer Questioning is an activity in which students question each other about the content they are learning using higher-order, open-ended question stems. The questions are used to promote thinking and generate focused discussions in small groups.

The students were in the class room. First the students were taught with the concepts of SONET/SDH Rings and Networks. Then the students were allowed to discuss the topics among their friends. Then the students were encouraged to ask questions to each other by which the concepts are learnt easily and more quickly.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO9	PO10	PO12	PSO3
CO5	3	2	3	3	2	2	1	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9	PSO3
	2	3
Justification for correlation	Reciprocal Peer Questioning activity on SONET/SDH networks helps the students to present results as a team, with smooth integration of contributions from all individual efforts.	The knowledge of SONET/SDH optical architecture, WDM, Passive WDM components is useful in designing Communication systems.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

The effectiveness of the activity was felt when the students were asked the questions on the topic, they were able to answer easily. They also understood the concepts more easily and able to remember while writing the test.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The students were more interested in learning the concepts if they are asking questions and sharing the concepts through their friends. They clarified their doubts with the neighboring students and the faculty. They learnt the concepts and asked more doubts in the topic and learnt many things. Outcome attainment has also been increased since the students' performance in the test is also good.

❖ ***Challenges faced in implementation:***

Basically some time is needed to explain the basic concepts of SONET/SDH to the students and also to discuss among the students.

References:

Textbooks:

1. P Chakrabarti, "Optical Fiber Communication, McGraw Hill Education (India) Private Limited, 2016 (UNIT I, II, III)
2. Gerd Keiser, "Optical Fiber communication" MC Graw -Hill international, 4th Edition, 2010

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ACTIVE LEARNING METHODS

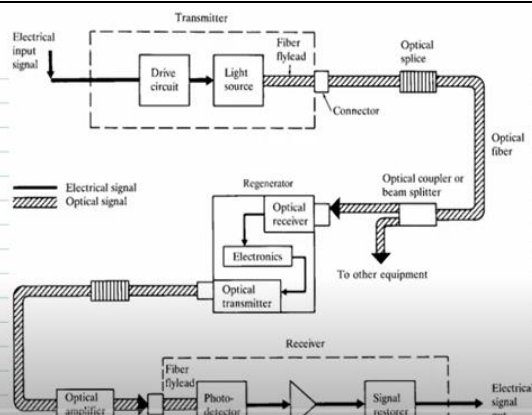
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Active Learning Method Execution

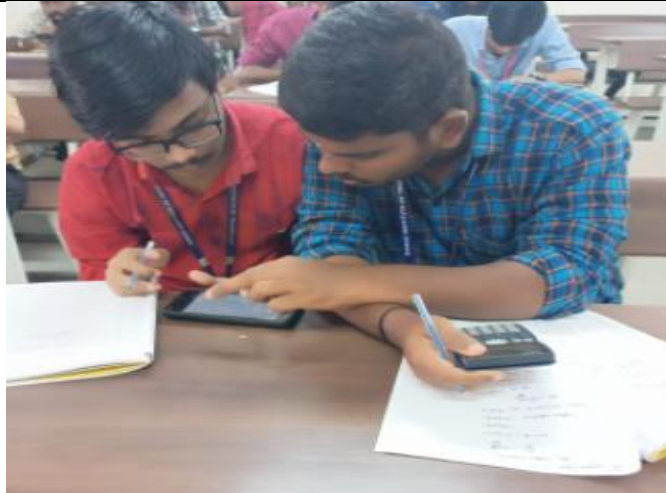
Unit I

Evolution of fiber optic system, Element of an optical Fiber Transmission link- Video

**Elements of
optical fiber
link**



Optical modes and configurations - Brainstorming



Modes in planar wave guide – Think Pair Share



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<u>Active Learning Method Execution</u>	
<u>Unit II</u>	
Evolution of fiber optic system, Element of an optical Fiber Transmission link- Video	
Bending losses, Core and Cladding losses - Concept Test	
	
Intra modal dispersion, Material dispersion, Waveguide dispersion – Quiz	
	

Dispersion optimization of single mode fiber – Mind Map



Dispersion shifted

Reduction in the fibre core diameter with an increase in the relative or fractional index difference to create dispersion is known as dispersion shifted fibre.

Dispersion biased

Fibre which relaxes the spectral properties on optical sources and allow bandwidth multiplication are known as dispersion biased fibre.

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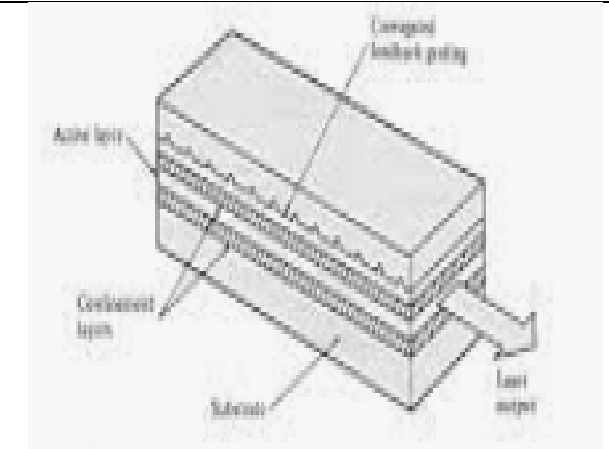
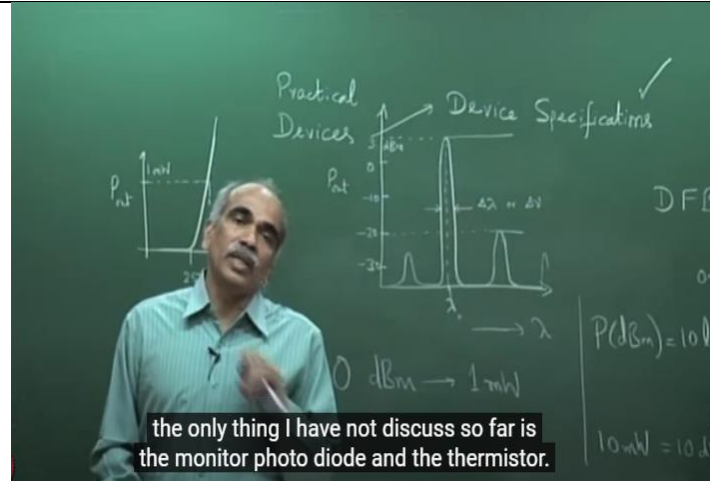
Active Learning Method Execution

Unit III

LED-LED structures surface emitting LED, Edge emitting LED – Class Poll



Laser Diodes, Modes and Threshold conditions – NPTEL Video



Laser structures and radiation patterns, Single mode laser, External modulation-temperature effort – In-Class Teams



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<u>Active Learning Method Execution</u>	
<u>Unit IV</u>	
Introduction to Fundamental Receiver operation, Preamplifiers – NPTEL Video	
	Lecture - 1 Introduction to Communicati... Watch la 
Optical power measurement: Attenuation measurement, Dispersion measurement – Flipped Class	
	

LED Coupling to Single Mode Fibers, Fiber Splicing, Optical Fiber connectors, Revision – Open Book Test (Library)



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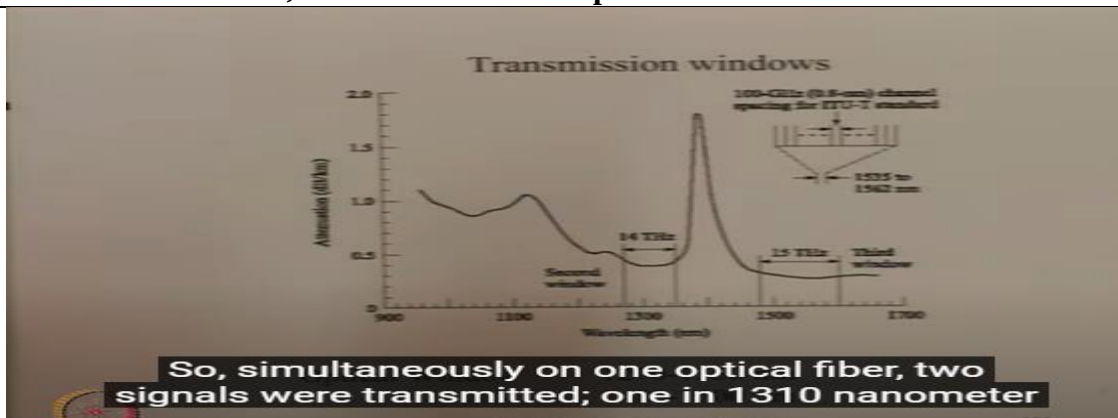
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Active Learning Method Execution

Unit V

WDM, Passive DWDM Components – NPTEL Video



SONET/SDH Rings and Networks – Reciprocal Peer Questioning

